

Simple linear regression

- The sample: $\{(x_i, y_i)\}_{i=1}^n$

independent
variable

dependent
variable

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon$$

$$\varepsilon \sim N(0, \sigma^2)$$

random deviation or random error

Multiple linear regression with k independent variables. $k > 1$

- The sample: $\{(x_{1i}, x_{2i}, \dots, x_{ki}, y_i)\}_{i=1}^n$



independent
variables

dependent
variable

$$Y = \underbrace{\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_k X_k}_{\text{True regression line}} + \varepsilon \quad \varepsilon \sim N(0, \sigma^2)$$

True regression line

random deviation or random error

Summary of Multiple LR

The sample:

$$\{(x_{1i}, x_{2i}, \dots, x_{ki}, y_i)\}_{i=1}^n$$

$$Y = \underbrace{\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_k X_k}_{\text{True regression line}} + \varepsilon \quad \varepsilon \sim N(0, \sigma^2)$$

- Significance of LR model
 - $H_0: \beta_i = 0$ for all i
 - $H_a: \beta_i \neq 0$ for at least one i
 - F-test for the whole model
 - t-tests for individual independent variables. $H_0: \beta_i = 0$
 $H_a: \beta_i \neq 0$
- Percentage of variability explained by the LR model
 - R^2 : no reference to df
 - Adjusted R^2 : adjusted according to df so as to minimize MSE

Subjects

- Problems: Confounding and Collinearity
- Model selection:
 - Principle of Parsimony: “A model should contain the smallest number of variables necessary to fit the data.”

$$H_0: \beta_{k+1} = \cdots = \beta_p = 0$$

- Best subsets:
$$f = \frac{(SSE_{reduced} - SSE_{full})/(p - k)}{SSE_{full}/(n - p - 1)} \sim F_{p-k, n-(p+1)}$$
 - Full vs reduced models
 - Stepwise regression (partial F-tests): Forward selection & Backward elimination

How to choose the best regression model?

- Criteria

- coefficient of determination, R^2
- Higher adjusted coefficient of multiple determination:

$$r_a^2 = 1 - \frac{SSE/(n - k)}{SST/(n - 1)}$$

- Equivalently lower MSE
- Lower Mallows' C_p

Problems in multiple regression

1. Multicollinearity occurs when two or more of the regressors are “highly” correlated. Then the estimates of the corresponding β 's have high variances.

To detect multicollinearity:

1. Inspection of sample correlations between x_j and x_k for all j, k . If sample correlation $|r_{jk}| \approx 1$: suspect multicollinearity between x_j & x_k if more than 2 variables are correlated. this may not work.
2. Let x_j be one variable that is suspected to have high correlation with one or more of the other x_k .

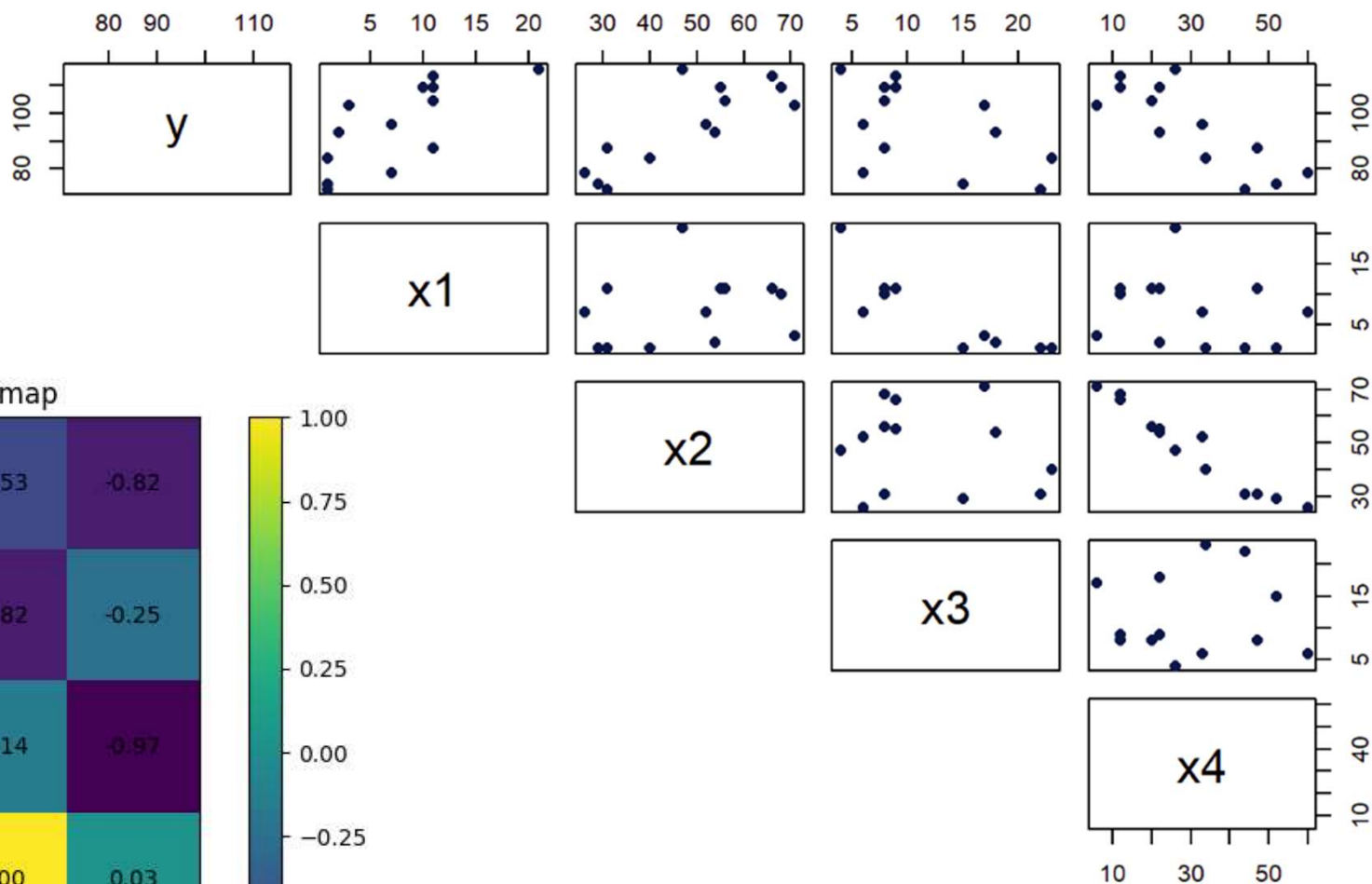
Regression of x_j on x_k : If R^2 is “high” multicollinearity.

3. F-test for regression is significant, but individual tests on β_i are not. Then suspect multicollinearity.
2. Influential observations in regression (problem for simple regression as well).
3. Autocorrelation (simple regression as well).

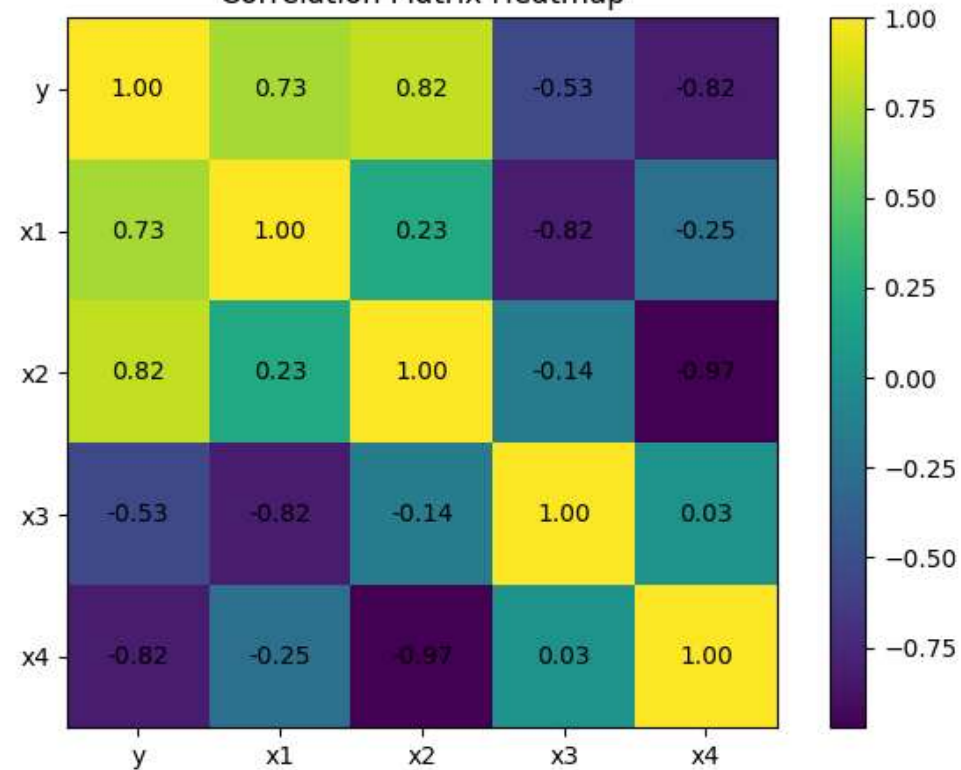
Example: Stepwise regression

- Let's learn how the stepwise regression procedure works by considering a data set that concerns the hardening of cement. Sounds interesting, eh? In particular, the researchers were interested in learning how the composition of the cement affected the heat that evolved during the hardening of the cement. Therefore, they measured and recorded the following data ([Cement dataset](#)) on 13 batches of cement:
 - Response, y : heat evolved in calories during the hardening of cement on a per gram basis
 - Predictor, x_1 : % of tricalcium aluminate
 - Predictor, x_2 : % of tricalcium silicate
 - Predictor, x_3 : % of tetracalcium aluminato ferrite
 - Predictor, x_4 : % of dicalcium silicate

Matrix Plot & Correlation Matrix



Correlation Matrix Heatmap



Stepwise regression

- Set: $\alpha_{\text{in}}=0.15$ & $\alpha_{\text{out}}=0.15$
- Step 1:
 - Build all models $Y = \beta_0 + \beta_k x_k + \varepsilon, k = 1, \dots, 4$
 - Find the predictor (independent variable) with the smallest t -test P -value, say .
 - If this is more than α_{in} , stop. Otherwise, keep this predictor x_p .

Constant	81.479	4.927	16.54	0.000
x1	1.8687	0.5264	3.55	0.005

Predictor	Coef	SE Coef	T	P
Constant	57.424	8.491	6.76	0.000
x2	0.7891	0.1684	4.69	0.001

Predictor	Coef	SE Coef	T	P
Constant	110.203	7.948	13.87	0.000
x3	-1.2558	0.5984	-2.10	0.060

Predictor	Coef	SE Coef	T	P
Constant	117.568	5.262	22.34	0.000
x4	-0.7382	0.1546	-4.77	0.001

Which predictor?

$$\alpha_{\text{in}}=0.15 \text{ \& } \alpha_{\text{out}}=0.15$$

Stepwise regression

- Step 2:
 - Build all models $Y = \beta_0 + \beta_4 x_4 + \beta_k x_k + \varepsilon, k = 1, \dots, 3$
 - Find the predictor (independent variable) with the smallest t -test P -value, say .
 - If this is more than α_{in} , stop. Otherwise, keep this predictor x_p .

Predictor	Coef	SE Coef	T	P
Constant	103.097	2.124	48.54	0.000
x4	-0.61395	0.04864	-12.62	0.000
x1	1.4400	0.1384	10.40	0.000

Predictor	Coef	SE Coef	T	P
Constant	94.16	56.63	1.66	0.127
x4	-0.4569	0.6960	-0.66	0.526
x2	0.3109	0.7486	0.42	0.687

Predictor	Coef	SE Coef	T	P
Constant	131.282	3.275	40.09	0.000
x4	-0.72460	0.07233	-10.02	0.000
x3	-1.1999	0.1890	-6.35	0.000

Which predictor(s)?
In or out?

$$\alpha_{\text{in}}=0.15 \text{ \& } \alpha_{\text{out}}=0.15$$

Stepwise regression

- Step 3:
 - Build all models $Y = \beta_0 + \beta_4 x_4 + \beta_1 x_1 + \beta_k x_k + \varepsilon, k = 2,3$
 - Find the predictor (independent variable) with the smallest t -test P -value, say .
 - If this is more than α_{in} , stop. Otherwise, keep this predictor x_p .

Predictor	Coef	SE Coef	T	P
Constant	71.65	14.14	5.07	0.001
x4	-0.2365	0.1733	-1.37	0.205
x1	1.4519	0.1170	12.41	0.000
x2	0.4161	0.1856	2.24	0.052

Predictor	Coef	SE Coef	T	P
Constant	111.684	4.562	24.48	0.000
x4	-0.64280	0.04454	-14.43	0.000
x1	1.0519	0.2237	4.70	0.001
x3	-0.4100	0.1992	-2.06	0.070

Which predictor(s)?
In or out?

$$\alpha_{\text{in}}=0.15 \text{ \& } \alpha_{\text{out}}=0.15$$

Stepwise regression

- Step 4:
 - Build all models $Y = \beta_0 + \beta_2 x_2 + \beta_1 x_1 + \beta_k x_k + \varepsilon, k = 3, 4$
 - Find the predictor (independent variable) with the smallest t -test P -value, say .
 - If this is more than α_{in} , stop. Otherwise, keep this predictor x_p .

Predictor	Coef	SE Coef	T	P
Constant	71.65	14.14	5.07	0.001
x1	1.4519	0.1170	12.41	0.000
x2	0.4161	0.1856	2.24	0.052
x4	-0.2365	0.1733	-1.37	0.205

Predictor	Coef	SE Coef	T	P
Constant	48.194	3.913	12.32	0.000
x1	1.6959	0.2046	8.29	0.000
x2	0.65691	0.04423	14.85	0.000
x3	0.2500	0.1847	1.35	0.209

Which predictor(s)?
In or out?

$$\alpha_{\text{in}}=0.15 \text{ \& } \alpha_{\text{out}}=0.15$$

Stepwise regression: Summary

Terms	-----Step 1-----		-----Step 2-----		-----Step 3-----		-----Step 4-----	
	Coef	P	Coef	P	Coef	P	Coef	P
Constant	117.57		103.10		71.6		52.58	
x4	-0.738	0.001	-0.6140	0.000	-0.237	0.205		
x1			1.440	0.000	1.452	0.000	1.468	0.000
x2					0.416	0.052	0.6623	0.000
S		8.96390		2.73427		2.30874		2.40634
R-sq		67.45%		97.25%		98.23%		97.44%
R-sq(adj)		64.50%		96.70%		97.64%		97.44%
Mallows' Cp		138.73		5.50		3.02		2.68

$$\alpha_{\text{in}}=0.15 \text{ \& } \alpha_{\text{out}}=0.15$$

Stepwise regression: Final model

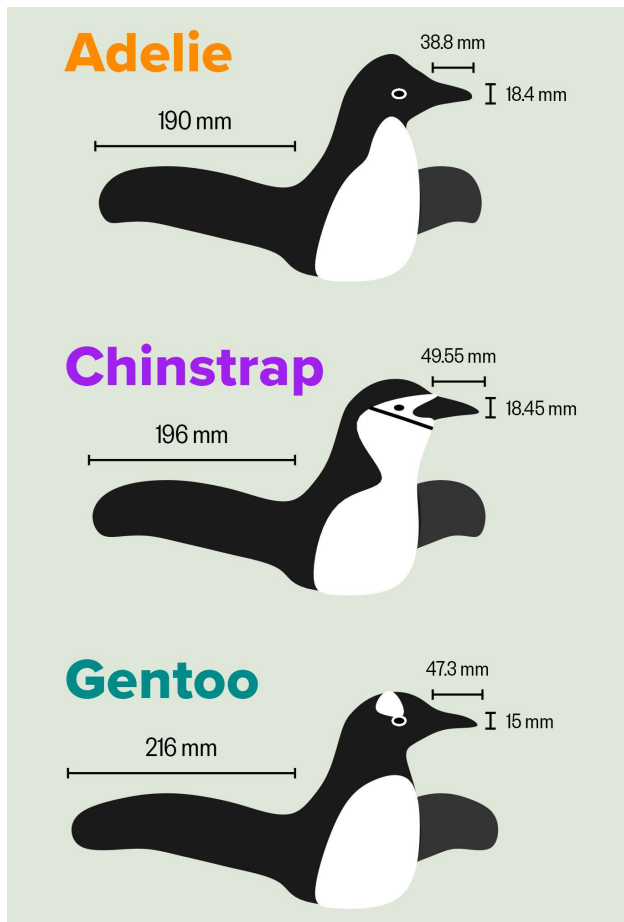
Predictor	Coef	SE Coef	T	P
Constant	52.577	2.286	23.00	0.000
x1	1.4683	0.1213	12.10	0.000
x2	0.66225	0.04585	14.44	0.000

S 2.40634

R-sq 97.44%

R-sq(adj) 97.44%

Mallows' Cp 2.68

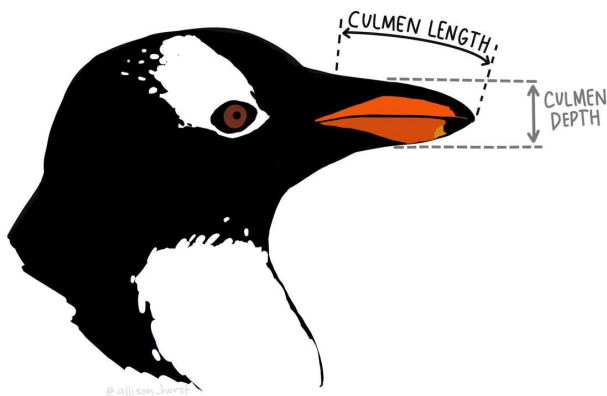


Penguins:

Stepwise regression

- Data collected from three species (Adelie, Chinstrap, Gentoo) and three islands (Biscoe, Dream, Torgersen)
- Data: Body mass, flipper_length_mm, culmen_length_mm, culmen_depth_mm
- Aim: to predict Body mass

CULMEN: RIDGE ALONG THE TOP PART OF A BIRD'S BILL



How to represent the species and islands?

- Species: Adelie, Chinstrap, Gentoo
 - species_Chinstrap: 0 or 1
 - species_Gentoo: 0 or 1
- Islands: Biscoe, Dream, Torgersen
 - island_Dream: 0 or 1
 - island_Biscoe: 0 or 1

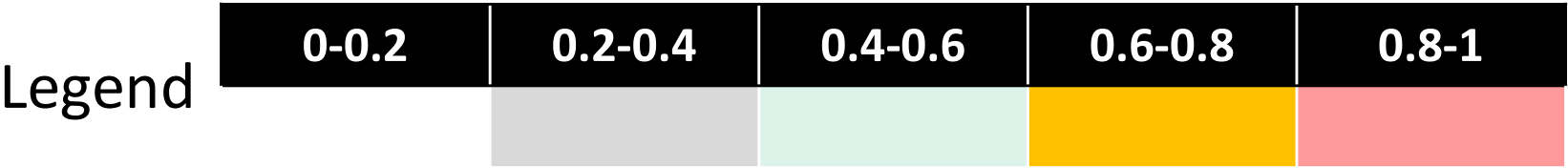
Descriptive Statistics

Dependent variable: body_mass_g

Variable	Mean	StDev	Min	Q1	Median	Q3	Max	Skewness
body_mass_g	4207.06	805.22	2700	3550	4050	4775	6300	0.472
culmen_length_mm	43.99	5.47	32.1	39.5	44.5	48.6	59.6	0.045
culmen_depth_mm	17.17	1.97	13.1	15.6	17.3	18.7	21.5	-0.150
flipper_length_mm	200.97	14.02	172	190	197	213	231	0.360
species_Chinstrap	0.204							
species_Gentoo	0.357							
island_Dream	0.369							
island_Biscoe	0.489							

Correlation Matrix

	body_mass	culmen_len	culmen_dep	flipper	Chinstrap	Gentoo	Dream	Biscoe
body_mass	1.000	0.589	-0.472	0.873	-0.299	0.821	-0.465	0.624
culmen_length	0.589	1.000	-0.229	0.653	0.449	0.488	0.032	0.225
culmen_depth	-0.472	-0.229	1.000	-0.578	0.324	-0.822	0.457	-0.626
flipper_length	0.873	0.653	-0.578	1.000	-0.186	0.867	-0.425	0.601
species_Chinstrap	-0.299	0.449	0.324	-0.186	1.000	-0.378	0.662	-0.496
species_Gentoo	0.821	0.488	-0.822	0.867	-0.378	1.000	-0.571	0.762
island_Dream	-0.465	0.032	0.457	-0.425	0.662	-0.571	1.000	-0.749
island_Biscoe	0.624	0.225	-0.626	0.601	-0.496	0.762	-0.749	1.000



Step 1 – Choosing the first variable

$$Y = \beta_0 + \beta_k x_k + \varepsilon, k = 1, \dots, 7$$

Label	Variable	R ²	p-value
1	flipper_length_mm	0.7621	3.1 * 10 ⁻¹⁰⁵
2	species_Gentoo	0.6743	1.23 * 10 ⁻⁸²
3	island_Biscoe	0.3890	2.7 * 10 ⁻³⁷
4	culmen_length_mm	0.3475	1.5 * 10 ⁻³²
5	culmen_depth_mm	0.2228	7.0 * 10 ⁻²⁰
6	island_Dream	0.2159	3.1 * 10 ⁻¹⁹
7	species_Chinstrap	0.0892	2.8 * 10 ⁻⁸

Step 1 - Model

$$Y = \beta_0 + \beta_1 x_1 + \varepsilon$$

Label	Variable
1	flipper_length_mm
2	species_Gentoo
3	island_Biscoe
4	culmen_length_mm
5	culmen_depth_mm
6	island_Dream
7	species_Chinstrap

Variable	Estimate
Intercept	-5872.09
flipper_length_mm	50.15

Source	df	SS	MS	F
Model	1	164,047,703	164,047,703	1060.30
Residual	331	51,211,963	154,719	
Total	332	215,259,666		

Step 2 – Choosing the second variable

$$Y = \beta_0 + \beta_1 x_1 + \beta_k x_k + \varepsilon, k = 2, \dots, 7$$

Label	Added Variable	Adj-R ²	p-value
7	species_Chinstrap	0.7800	$1.41 * 10^{-7}$
2	species_Gentoo	0.7775	$9.71 * 10^{-7}$
3	island_Biscoe	0.7761	$2.81 * 10^{-6}$
6	island_Dream	0.7713	$1.05 * 10^{-4}$
5	culmen_depth_mm	0.7622	0.139
4	culmen_length_mm	0.7613	0.342

Step 2 - Model

$$Y = \beta_0 + \beta_1 x_1 + \beta_7 x_7$$

Label	Variable
1	flipper_length_mm
2	species_Gentoo
3	island_Biscoe
4	culmen_length_mm
5	culmen_depth_mm
6	island_Dream
7	species_Chinstrap

Variable	Estimate
Intercept	-5511.63
flipper_length_mm	48.65
species_Chinstrap	-281.18

Source	df	SS	MS	F
Model	2	168,177,890	84,088,945	589.39
Residual	330	47,081,776	142,672	

Step 3 – Choosing the third variable

$$Y = \beta_0 + \beta_1 x_1 + \beta_7 x_7 + \beta_k x_k + \varepsilon, k = 2, \dots, 6$$

Label	Variable	Adj-R ²	p-value
4	culmen_length_mm	0.8218	4.97*10 ⁻¹⁷
5	culmen_depth_mm	0.7857	0.0018
2	species_Gentoo	0.7851	0.0031
3	island_Biscoe	0.7835	0.0124
6	island_Dream	0.7796	0.529

Step 3 - Model

$$Y = \beta_0 + \beta_1 x_1 + \beta_7 x_7 + \beta_4 x_4$$

Label	Variable
1	flipper_length_mm
2	species_Gentoo
3	island_Biscoe
4	culmen_length_mm
5	culmen_depth_mm
6	island_Dream
7	species_Chinstrap

Variable	Estimate
Intercept	-4423.71
flipper_length_mm	30.12
species_Chinstrap	-779.46
culmen_length_mm	62.22

Source	df	SS	MS	F
Model	3	177,253,617	59,084,539	511.47
Residual	329	38,006,049	115,520	

Step 4 – Choosing the fourth variable

$$Y = \beta_0 + \beta_1 x_1 + \beta_7 x_7 + \beta_4 x_4 + \beta_k x_k + \varepsilon, k = 2, 3, 5, 6$$

Label	Variable	Adj-R ²	p-value
5	culmen_depth_mm	0.8261	0.0029
2	species_Gentoo	0.8222	0.2051
3	island_Biscoe	0.8221	0.2188
6	island_Dream	0.8213	0.8259

Step 4 - Model

$$Y = \beta_0 + \beta_1 x_1 + \beta_7 x_7 + \beta_4 x_4 + \beta_5 x_5$$

Label	Variable
1	flipper_length_mm
2	species_Gentoo
3	island_Biscoe
4	culmen_length_mm
5	culmen_depth_mm
6	island_Dream
7	species_Chinstrap

Variable	Estimate
Intercept	-5592.15
flipper_length_mm	33.16
species_Chinstrap	-809.30
culmen_length_mm	61.07
culmen_depth_mm	35.79

Source	df	SS	MS	F
Model	4	178,269,723	44,567,431	395.19
Residual	328	36,989,942	112,774	

Step 5 – Choosing the fifth variable

$$Y = \beta_0 + \beta_1 x_1 + \beta_7 x_7 + \beta_4 x_4 + \beta_5 x_5 + \beta_k x_k + \varepsilon, k = 2, 3, 6$$

Label	Variable	Adj-R ²	p-value
2	species_Gentoo	0.8472	4.74 *10 ⁻¹¹
3	island_Biscoe	0.8289	0.0112
6	island_Dream	0.8256	0.8000

Step 5 - Model

$$Y = \beta_0 + \beta_1 x_1 + \beta_7 x_7 + \beta_4 x_4 + \beta_5 x_5 + \beta_2 x_2$$

Label	Variable
1	flipper_length_mm
2	species_Gentoo
3	island_Biscoe
4	culmen_length_mm
5	culmen_depth_mm
6	island_Dream
7	species_Chinstrap

Variable	Estimate
Intercept	-4282.08
flipper_length_mm	20.23
species_Chinstrap	-496.76
culmen_length_mm	39.72
culmen_depth_mm	141.77
species_Gentoo	965.20

Source	df	SS	MS	F
Model	5	182,861,995	36,572,399	369.14
Residual	327	32,397,671	99,075	

Step 6 – Choosing the sixth variable

$$Y = \beta_0 + \beta_1 x_1 + \beta_7 x_7 + \beta_4 x_4 + \beta_5 x_5 + \beta_2 x_2 + \beta_k x_k + \varepsilon, k = 3, 6$$

Label	Variable	Adj-R ²	p-value
3	island_Biscoe	0.8469	0.6009
6	island_Dream	0.8470	0.4372

Summary of models with respect to adjusted R^2

Step	Model	Adjusted R^2
Step 1	flipper_length	0.7614
Step 2	+ species_Chinstrap	0.7800
Step 3	+ culmen_length	0.8218
Step 4	+ culmen_depth	0.8261
Step 5	+ species_Gentoo	0.8472

Final Model

Label	Variable
1	flipper_length_mm
2	species_Gentoo
3	island_Biscoe
4	culmen_length_mm
5	culmen_depth_mm
6	island_Dream
7	species_Chinstrap

$$Y = \beta_0 + \beta_1 x_1 + \beta_7 x_7 + \beta_4 x_4 + \beta_5 x_5 + \beta_2 x_2 + \beta_k x_k + \varepsilon$$

Source	df	SS	MS	F
Model	5	182,861,995	36,572,399	369.14
Residual	327	32,397,671	99,075	

Variable	Estimate	p-value
Intercept	-4282.08	$3.33 * 10^{-16}$
flipper_length_mm	20.23	$3.98 * 10^{-10}$
species_Chinstrap	-496.76	$4.59 * 10^{-9}$
culmen_length_mm	39.72	$7.85 * 10^{-8}$
culmen_depth_mm	141.77	$1.17 * 10^{-12}$
species_Gentoo	965.20	$4.74 * 10^{-11}$

Adjusted R²=0.8472